

Ecohydrologic Controls on Vegetation Cooling in Urban Arid Environments

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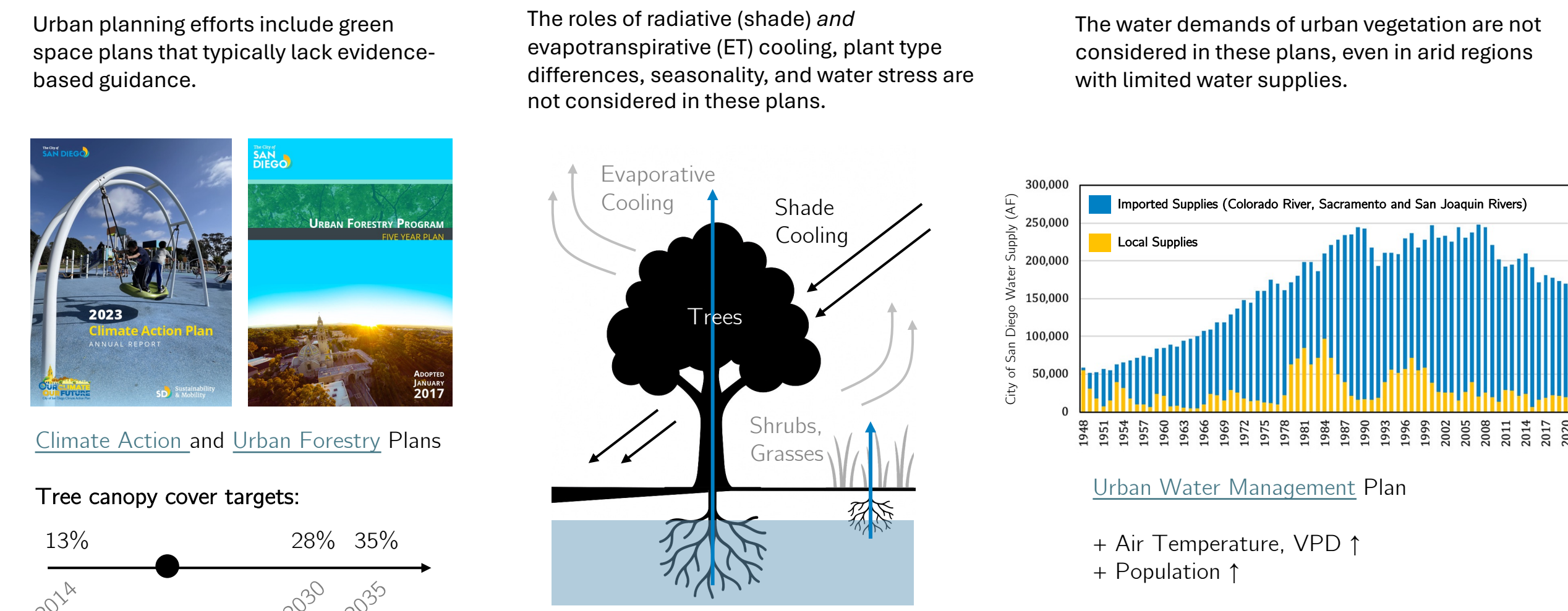
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Introduction

Why examine ecohydrologic (plant-water interaction) controls on vegetation cooling (urban greening for heat adaptation) in arid urban environments (southern California)?

Urban greening is promoted as a remedy for urban heat, yet its cooling effectiveness and water requirements remain poorly constrained, particularly in arid regions. This work focuses on hydroclimatic and ecohydrologic controls on green space cooling in arid urban and peri-urban environments. We integrate publicly available spatiotemporal datasets -- including land surface temperatures, vegetation greenness, evapotranspiration, soil moisture, and other meteorological and geophysical observations -- in a case study of San Diego County, California, U.S.A. Using a framework grounded in surface energy balance theory and ecohydrologic process understanding, we apply a novel spatiotemporal sampling design to assess how vegetation, water, and climate interact to determine cooling and associated water demand outcomes.



Background

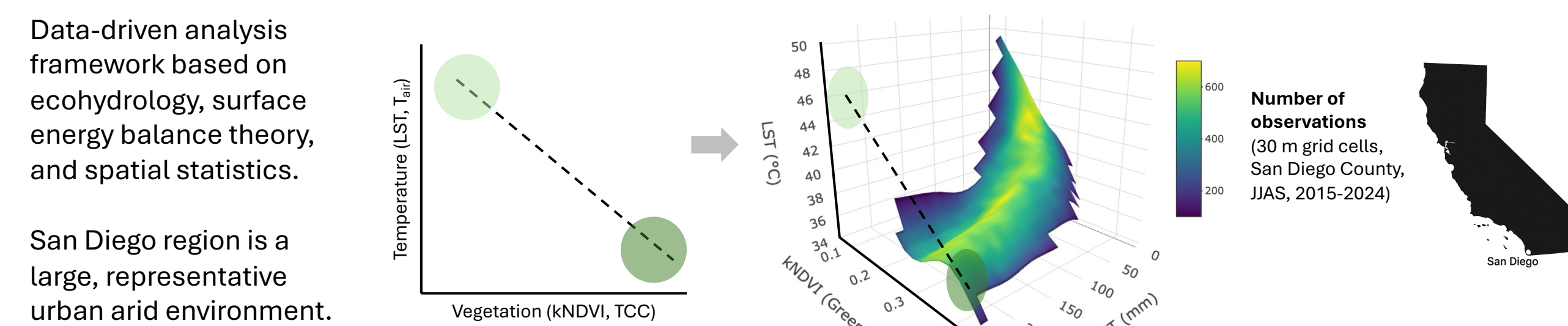
Research:

- Approach:** Estimate “Cooling Rates”, or the change (decrease) in Land Surface Temperature (LST) with a change (increase) in Canopy Cover (CC) or Leaf Area Index (LAI): $-\Delta LST / \Delta CC$.
- Findings:** Greener spaces or those with greater canopy cover are cooler and offer multiple co-benefits, and evapotranspirative cooling can dominate overall cooling in arid environments.

Research Gaps:

- Vegetation metrics that capture seasonality and water stress
- Use of fine-scale spatial and temporal data
- Place-based estimates of variability and uncertainty
- Water availability and constraints at local and regional scales
- Applications in arid regions, where:
 - Cooling may be primarily driven by ET, and
 - Cooling may be sensitive to vegetation type, water stress, and placement

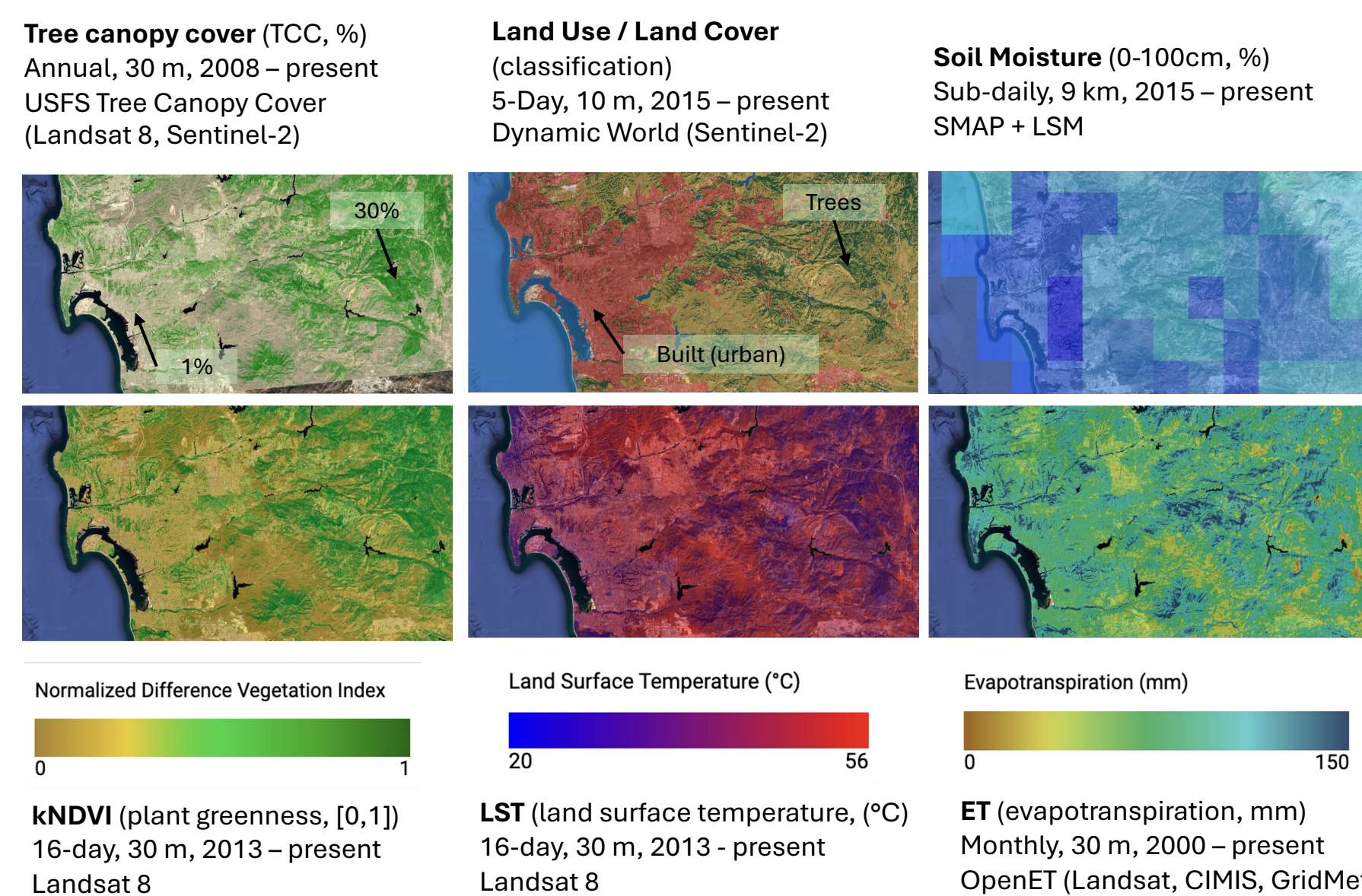
Methods: Conceptual Framework



Methods: Data

Figure 1. Select variables, data sources, spatial and temporal resolutions, and temporal coverage.

Data were acquired and pre-processed using Google Earth Engine (earthengine.google.com). Additional data sources not shown include maximum and minimum air temperature, precipitation, wind speed, and vapor pressure deficit from GridMET (4 km, daily), and elevation from NASA SRTM (30 m).



Methods: Space-For-Time Sampling Design

- Generated monthly samples of 30 m resolution data** (100 per kNDVI x LULC stratum) from randomly-sampled point locations across San Diego County for each summer month (June–September) from 2016–2023. Samples include point locations, month-year identifiers, and co-located environmental variables. Sampling was limited to locations with average LULC classification probabilities > 50%, LST was filtered to the central 90% of its regional distribution, and kNDVI limited to 0–0.6 to achieve balanced sampling across the observed range of kNDVI.
- Performed a custom three-step spatial thinning procedure** to reduce spatial clustering and ensure balanced LULC representation, including grid-based thinning (within LULC class), distance-based thinning (across LULC classes), and random sampling (within LULC class) to ensure balanced LULC class representation.
 - A first sampling strategy (S1)** results in 4,000 spatially-unique points sampled once in time, 1,000 per LULC class, with each point separated by a minimum distance of 50 m.
 - A second sampling strategy (S2)** is like S1 but is constructed from repeated time observations at static (vs. variable) locations, which are also spatially thinned and subsampled for LULC balance (as in S1). This results in ~2,000 repeated time observations from ~200 locations (~50 locations per LULC).

Samples capture fine-scale relationships between temperature, vegetation, and water demand, which can be analyzed in aggregate at landscape scales relevant to local and regional planning.

Results: Exploratory Analysis

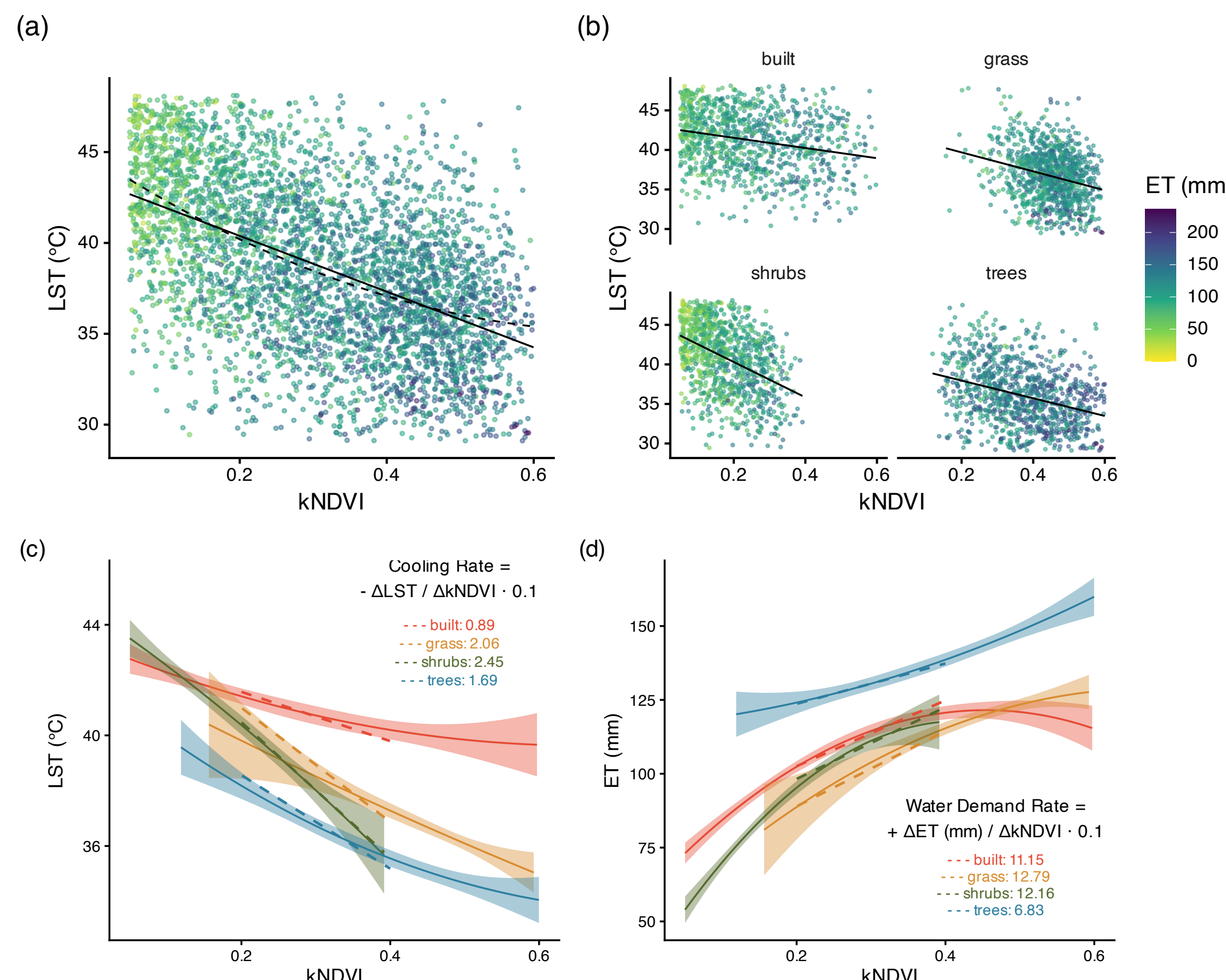


Figure 2. Relationships between land surface temperature (LST), greenness (kNDVI), and landscape water demand (ET). (a) Scatterplot of monthly mean LST (°C) versus kNDVI colored by monthly total evapotranspiration (ET, mm), overlaid with linear (solid line) and second-order polynomial (dashed line) fits. (b) Same as (a) but stratified by LULC class. (c, d) Nonlinear (second-order polynomial) relationships between kNDVI and (c) land surface temperature (LST, °C) and (d) evapotranspiration (ET, mm), fitted separately for each LULC class. Dashed lines and colors represent linear fits over a subset kNDVI range (0.2–0.4), used to estimate representative LULC-specific cooling rates ($-\Delta LST$ per 0.1 increase in kNDVI) and water-demand rates ($+\Delta ET$ per 0.1 increase in kNDVI). Data: S1.

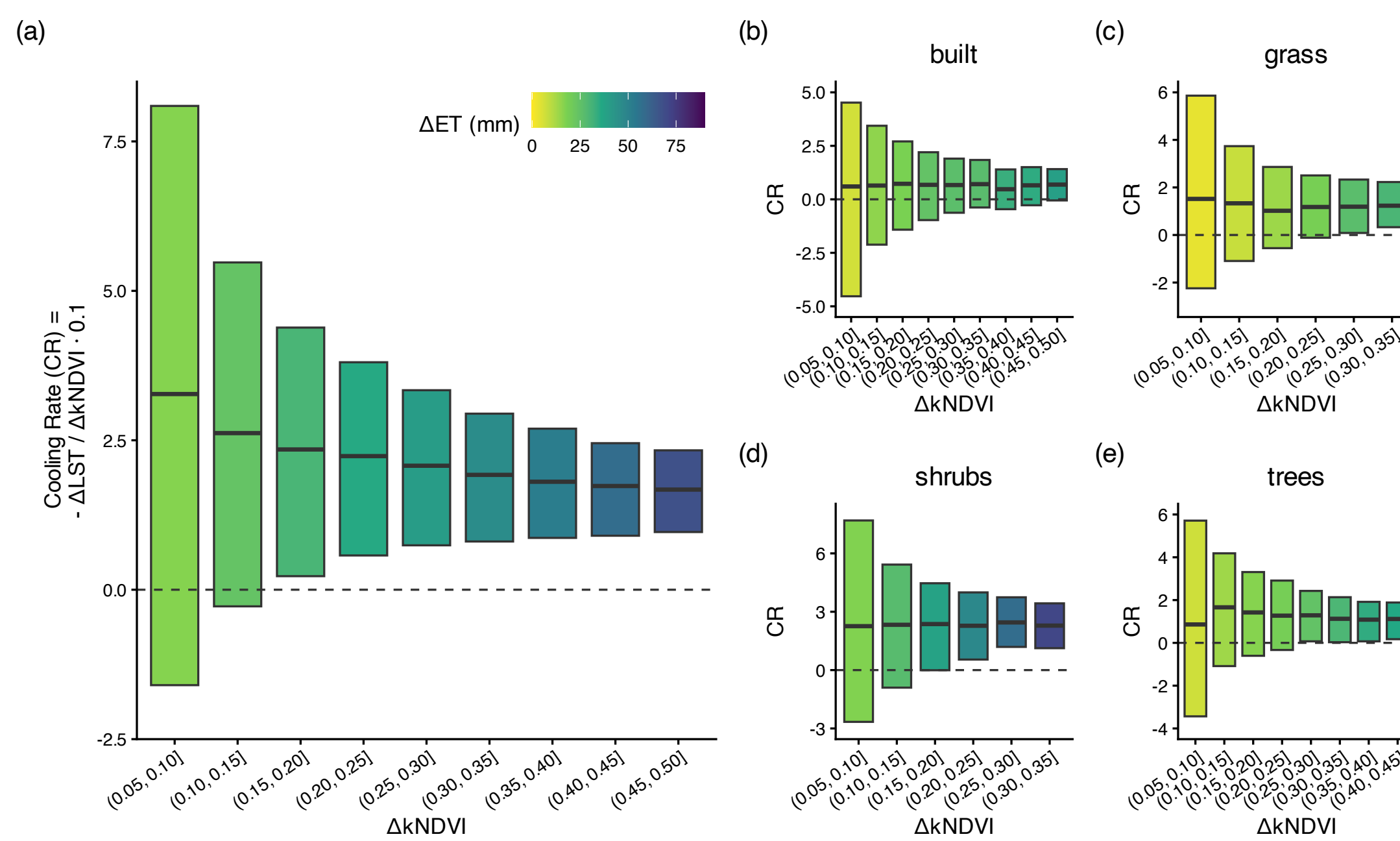


Figure 3. Cooling rate variability across increasing greenness ($\Delta kNDVI$). (a) Interquartile ranges (25th–75th percentiles) of cooling rates ($CR = -\Delta LST / \Delta kNDVI \cdot 0.1$) computed from pairs of sample points whose kNDVI difference ($\Delta kNDVI$) falls within the labeled interval. Boxes are colored by the mean associated change in evapotranspiration (ΔET , mm) for each interval, and horizontal dashed lines indicate no cooling. (b–e) Same as (a), shown separately for each LULC class. Axis limits vary by panel. Data: S1.

Methods: Inferential Statistical Analysis

We quantify LULC-specific cooling and water-demand responses to greenness while accounting for spatial dependence and sampling uncertainty using a custom **spatial bootstrap with repeated Generalized Additive Model (GAM)** fits.

$$LST_i = \beta_0 + \beta_{LULC(i)} s_i(kNDVI_i) + s(lon_i, lat_i) + b_{month(i)} + b_{year(i)} + \epsilon_i$$

i : sampled location
 $\beta_{LULC(i)}$: fixed effect for LULC class i
 $s_i(kNDVI_i)$: LULC-specific smooth of kNDVI
 $s(lon_i, lat_i)$: 2D spatial smooth
 $b_{month(i)}, b_{year(i)}$: month and year random effects
 $\epsilon_i \sim N(0, \sigma^2)$: residual error

Cooling Rates: We iteratively perform spatial bootstrap resampling of location points using the S1 sampling scheme (500 iterations). Within each iteration, we fit a GAM that estimates LULC-specific smooth responses of LST to kNDVI, while controlling for spatial structure and monthly and interannual variability.

The smooth term $s_i(kNDVI_i)$ represents the LULC-specific relationship between greenness and LST. We estimate cooling rates as the derivative of this smooth, evaluated across the kNDVI range.

For each GAM refit on spatially resampled data, we estimated within-model GAM uncertainty (σ^2_{GAM}) as the mean model-based variance (squared standard error) of partial smooth estimates across iterations, and between-iteration (sampling) uncertainty (σ^2_{sample}) as the variance of partial smooth estimates across iterations. We combined these to estimate total uncertainty:

$$\sigma^2_{total}(kNDVI) = \sigma^2_{GAM}(kNDVI) + \sigma^2_{sample}(kNDVI)$$

σ^2_{GAM} represents mean within-model GAM uncertainty
 σ^2_{sample} represents between-iteration (sampling) uncertainty

Water demand Rates: In the above GAM structure, the outcome variable is instead ET_i , and the smooth term $s_i(kNDVI_i)$ represents the LULC-specific relationship between kNDVI and ET. We estimate water demand rates as the derivative of this smooth, evaluated across the kNDVI range.

Results: Inferential Statistical Analysis

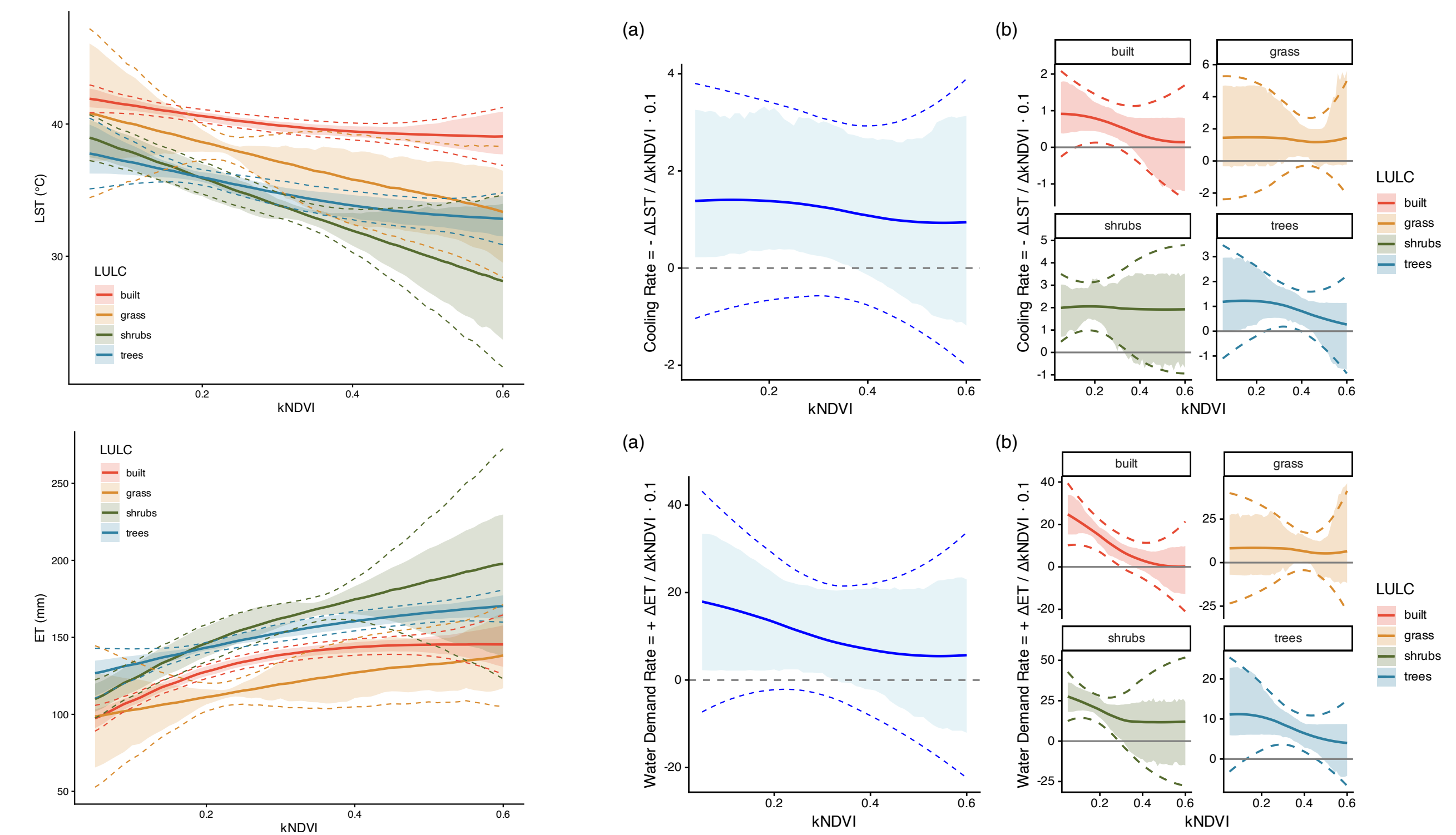
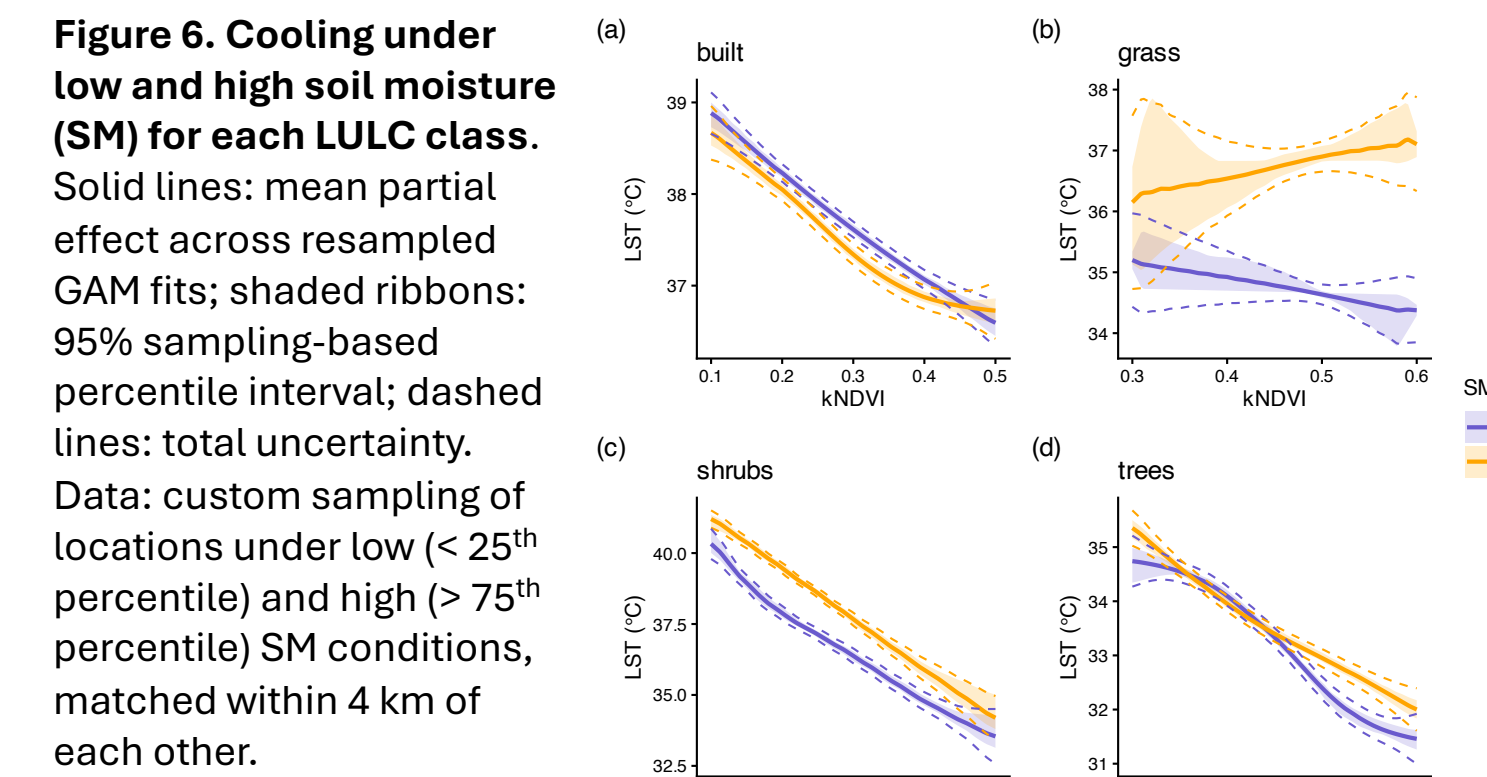


Figure 4. Spatial bootstrap estimates of the relationship between kNDVI and LST (top) or ET (bottom) (GAM-derived partial effects). Solid lines show the mean across 500 resampled GAM fits. Shaded ribbons show the 95% sampling-based percentile interval, and dashed lines represent total uncertainty ($\sigma^2_{total}(kNDVI)$) combining within- and between-iteration variance.

Figure 5. Spatial bootstrap cooling (top) and water demand (bottom) rates at different levels of greenness. (a) Cooling rates (top) or water demand rates (bottom) estimated as the derivative of the GAM smooth effects (Figure 4) for all LULC classes combined (a) or within LULC class (b). Solid lines, shaded ribbons, and dashed lines are as in Figure 4.

Results: Further Details

- Role of soil moisture is hard to establish (due to low spatial resolution), but stratification across low and high soil moisture levels suggest more cooling under higher soil moisture for shrubs (trees), especially at lower (higher) levels of greenness (Figure 6).
- Monte Carlo GAM cooling and water demand rates (Figures 4, 5) are attenuated relative to EDA rates (Figure 2).
- S2 samples generate similar results, with the cooling rates for trees (grass) being less (more) uncertain than with S1 samples.
- GAM models fit with meteorologic and geophysical variables in place of 2D spatial smooths generate similar results.
- All Monte Carlo GAM results demonstrate no residual spatial autocorrelation according to Moran’s I tests.
- Equivalently-structured OLS models yield interpretation similar to GAM, but with spatial autocorrelation unaddressed.
- Cooling rates according to air temperature (vs LST) are significantly attenuated.



Conclusions:

- Shrub landscapes show greater cooling rates than trees, specifically at lower levels of kNDVI.
- Shrub landscapes also show greater water demand rates than trees, but comparable overall water demand at low and moderate kNDVI.
- Grass provides cooling across all levels of kNDVI at relatively low levels of water demand, but that cooling is uncertain.
- Built environments are warmer across all kNDVI levels, and cooling rates are low, indicating potentially limited landscape-scale green space cooling potential in highly urbanized areas.